

IMAGINARY ELECTRIC COMPANY

Power Generation Division - Plant Integration

Alpilla 600 MW Combined Cycle Power Plant - 1 x IE-9H.02 multi-shaft power island

Division of Responsibility - Power Island (IEC) and Balance of Plant (EPC)

Document number	IEC-ALP-DOR-001
Revision	0
Date	2026-09-27
Status	Issued for EPC engineering (reference data)
Addressee	Istanbul EPC Muhendislik ve Taahhut A.S. (consortium leader) - EPC engineering
Contract	ALP-EPC-001 (Owner contract); consortium agreement IEPC-IEC

	Prepared	Checked	Approved
Function	IEC plant integration engineer	IEC lead engineer (discipline)	IEC project engineering manager
Signature / date	signed 2026-09-27	signed 2026-09-27	signed 2026-09-27

Rev	Date	Description
0	2026-09-27	Issued for EPC engineering (reference data)

CASE STUDY - Imaginary Electric, its products and all data are fictional.

1. Division of responsibility between IEC and the EPC

Design / Supply / Erection / Supervision (of erection) / Commissioning. 'IEC with EPC' = IEC leads, EPC provides labour, utilities and plant systems. Items not listed follow the consortium agreement: power island equipment by IEC (Owner contract items SC-006 to SC-010, SC-012 to SC-014, SC-029, SC-037, SC-054), everything else by EPC.

No	Item	Design	Supply	Erection	Supervision	Commissioning	Remarks
1.1	Gas turbine package incl. enclosure, lube oil, hydraulic and fuel skids	IEC	IEC	EPC	IEC	IEC with EPC	
1.2	GT inlet air filter house, anti-icing, silencer, inlet duct	IEC	IEC	EPC	IEC	IEC	evaporative cooler: space provision only (not included)
1.3	GT exhaust diffuser and expansion joint to HRSG	IEC	IEC	EPC	IEC	IEC	
1.4	Performance fuel gas heater and GT gas skid (from IF-01)	IEC	IEC	EPC	IEC	IEC	
1.5	Gas receiving and metering station, start-up gas heating, final filtration 3 um	EPC	EPC	EPC	-	EPC	IEC fuel gas requirements per DS-101
1.6	Liquid fuel skid, water injection skid, purge and recirculation (from IF-02/03)	IEC	IEC	EPC	IEC	IEC	
1.7	LDO unloading, storage, forwarding and treatment	EPC	EPC	EPC	-	EPC	
1.8	Compressor washing skid	IEC	IEC	EPC	IEC	IEC	demin water by EPC
1.9	GT enclosure fire suppression (CO2) and gas detection	IEC	IEC	EPC	IEC	IEC	fire alarm panel by EPC
2.1	HRSG pressure parts, casing, drums, platforms, CPH recirculation pumps	IEC	IEC	EPC	IEC	EPC with IEC	
2.2	SCR catalyst, ammonia injection grid and skid, CO catalyst	IEC	IEC	EPC	IEC	EPC with IEC	
2.3	Aqueous ammonia unloading, storage and forwarding	EPC	EPC	EPC	-	EPC	not in the Owner contract scope list
2.4	HRSG blowdown tanks, drain tanks, atmospheric flash tanks	EPC	EPC	EPC	-	EPC	IEC gives flows and conditions
2.5	Chemical dosing and steam/water sampling (SWAS)	EPC	EPC	EPC	-	EPC	IEC sample/dosing nozzles
2.6	Stack incl. silencer, CEMS, aviation lighting	EPC	EPC	EPC	-	EPC	65 m (ER-06.06)
3.1	Steam turbine with valves, lube/EHC oil, gland steam system, turning gear	IEC	IEC	EPC	IEC	IEC	
3.2	HP / IP / LP bypass valves with desuperheaters	IEC	IEC	EPC	IEC	IEC with EPC	loose supply
3.3	HP, cold reheat, hot reheat and LP steam piping HRSG - ST incl. supports	EPC	EPC	EPC	-	EPC	IEC: nozzle loads, pressure drop assumptions (PER-001 1)
3.4	Bypass piping and spray water piping	EPC	EPC	EPC	-	EPC	
3.5	ST drains, drain collection and flash tank	EPC	EPC	EPC	-	EPC	IEC drain list
4.1	Condenser incl. dump devices, hotwell, vacuum pumps	IEC	IEC	EPC	IEC	EPC with IEC	
4.2	Tube cleaning ball system and debris filters	EPC	EPC	EPC	-	EPC	required by ER-07.03
4.3	Circulating water system, intake, pumps, outfall, electro-chlorination	EPC	EPC	EPC	-	EPC	
4.4	Condensate pumps, boiler feed pumps, condensate polishing	EPC	EPC	EPC	-	EPC	IEC pressure assumptions PER-001
4.5	Closed cooling water system	EPC	EPC	EPC	-	EPC	IEC cooler demands REQ-001 5.1
5.1	GT and ST generators incl. excitation, neutral earthing	IEC	IEC	EPC	IEC	IEC	
5.2	Generator circuit breakers, isolated phase busducts	IEC	IEC	EPC	IEC	IEC	
5.3	GT and ST generator step-up transformers	IEC	IEC	EPC	IEC	IEC with EPC	
5.4	SFC and isolation transformer	IEC	IEC	EPC	IEC	IEC	
5.5	Generator and GSU protection panels	IEC	IEC	EPC	IEC	IEC with EPC	
5.6	H2 and CO2 bottle racks and supply piping	EPC	EPC	EPC	-	EPC	IEC gas control panel
5.7	Unit auxiliary transformers, MV/LV switchgear, UPS/DC, cabling	EPC	EPC	EPC	-	EPC	
5.8	380 kV connection GSU - GIS, GIS switchyard	EPC	EPC	EPC	-	EPC	
6.1	Turbine control system (TCS) for GT, ST, generators, HRSG protection (SIS)	IEC	IEC	EPC	IEC	IEC	stand-alone TCS with OPC UA gateway to the DCS
6.2	Plant DCS incl. HRSG and BOP control, operator stations	EPC	EPC	EPC	-	EPC	TCS HMI screens via gateway
6.3	CEMS	EPC	EPC	EPC	-	EPC	
6.4	Vibration and performance monitoring, remote diagnostics	IEC	IEC	EPC	IEC	IEC	secure link by EPC (IEC 62443)
7.1	Foundations of IEC equipment (GT, generators, ST, HRSG, condenser, GSUs)	EPC	EPC	EPC	-	-	IEC load data, anchor bolts and sole plates

No	Item	Design	Supply	Erection	Supervision	Commissioning	Remarks
7.2	Turbine hall(s), cranes, HRSG access structures beyond IEC platforms	EPC	EPC	EPC	-	-	
8.1	Transport of IEC equipment to site (incl. heavy transport, barge, SPMT)	-	IEC to port of discharge; EPC from port to foundation	-	-	-	barge landing TP-M1 by EPC
8.2	Performance test	EPC with IEC	EPC (test instruments)	-	IEC	EPC	ASME PTC 46
8.3	Training of Owner staff on IEC equipment	IEC	IEC	-	-	-	